

OPENSIFT EX280v4.18 Exam

Important instructions:

- utility.lab.example.com (172.25.250.14)
- master01.lab.example.com 172.25.250.12
- Server API URL: <https://api.ocp4.example.com:6443>

Questions Outline

- Configure the identity Providers for Openshift
- Configure Cluster permission
- Configure Project Permissions
- Create Groups and Configure Permissions
- Configure Quotas for Projects
- Configure and Deploy a secure route
- Scala the Application manually .
- Configure

1. Configure an identity provider

Configure your OpenShift cluster to use an **HTPasswd identity provider** with the following requirements:

- The name of the identity provider is **ex280-htpasswd**
- The name of the secret is **ex280-idp-secret**
- The user account **armstrong** is present and can log in with password **indiacore**
- The user account **collins** is present and can log in with password **verastar**
- The user account **aldrin** is present and can log in with password **moonhar**
- The user account **jobs** is present and can log in with password **eastiver**
- The user account **wozniak** is present and can log in with password **glimpse**

2. Configure cluster permission

Configure your OpenShift cluster to meet the following requirements:

- The user account **jobs** can perform cluster administration tasks
- The user account **wozniak** can create projects
- The user account **wozniak** cannot perform cluster administration tasks
- The user account **armstrong** cannot create projects
- The user account **kubeadmin** is not present.

3. Configure project permissions

Configure your OpenShift cluster to meet the following requirements:

The following projects exist:

- apollo
- manhattan
- gemini
- bluebook
- titan
- The user account **armstrong** is an administrator for project **apollo** and project **gemini**
- The user account **wozniak** can view project **titan** but not administer or delete it

4. Configure groups

Configure your OpenShift cluster to meet the following requirements:

- The user account **armstrong** is a member of the **commander** group
- The user account **collins** is a member of the **pilot** group
- The user account **aldrin** is a member of the **pilot** group
- Members of the **commander** group have edit permission in the **apollo** project
- Members of the **pilot** group have view permission in the **apollo** project

5. Configure quotas

Configure your OpenShift cluster to use quotas in the **manhattan** project with the following requirements:

- The name of the quota is **ex280-quota**
- The amount of memory consumed across all containers may not exceed **15Gi**
- The total amount of CPU consumed across all containers may not exceed **2 full cores**
- The maximum number of replication controllers does not exceed **3**
- The maximum number of pods does not exceed **3**
- The maximum number of services does not exceed **6**

6. Configure and deploy a secure route Deploy an application called oxcart securely in the project called area51

- The application has self-signed certificate available at `"/C=US/ST=NC/L=Raleigh/O=RedHat/OU=RHT/CN=oxcart.apps.ocp4.example.com"` Use `newcert` command to generate required tls objects, pass the subject as argument value
- The application should be reachable at the following url **`https://oxcart.apps.ocp4.example.com`**
- Application produces a valid Output

7. Deploy an application

Deploy the chart named **redhat-movie** in the project **aeti-service** from the repository:
<http://helm.domain6.example.com/charts/>

You may use the **telnet** or **nc** commands to validate the deployment.

8. Configure a service account

Configure a service account in the **apples** project to meet the following requirements:

- The name of the service account is **ex280sa**
- The service account allows pods to be run as **any available use**

9. Deploy an application

Deploy the application called **oranges** in the **apples** project so that the following conditions are true:

- The application uses the **ex280sa** service account
- No configuration components have been added or removed
- The application produces output.

10. Scale the Application manually

- Scale an application called **hydra** in the project called **lerna**. The hydra application should be scaled to five times

11. Configure a Secret

- Configure a secret in the **math** project and the name of the secret should be **magic**.
- The secret should have following key value pairs **ACCESS_STRING**:
ASDA142hfh-gfrhhueo-erfdk345v

12. Configure an application to use a secret

Configure the application called **red** in the **math** project with the following requirements:

- The application uses the secret previously created called **magic**
- The secret defines an environment variable with name **ACCESS_STRING**
- The application output no longer displays:
Sorry, application is not configured correctly

13. Configure a network policy

Configure a network policy using the **database** and **checker** projects with the following requirements:

- The database project has a network policy with the name db-allow-mysql-conn based on pod selector label network.openshift.io/policy-group
- Connections to the database project are restricted to deployments from the checker project
- The network policy is filtered by:
 - Project selector using the team=devsecops label
 - Pod selector using the deployment=web-mysql label
- The application can establish a connection to port 3306/TCP

You can check your work by examining the logs in the **checker** project.

14. Deploy an application

Deploy the application called **atlas** in the **mercury** project so that the following conditions are true:

- No configuration components have been added or removed
- The application produces output.

15. Install an operator

Install the **file-integrity** operator with the following requirements:

- The operator is installed in the **openshift-file-integrity** project
- The approval strategy is **Automatic**
- Cluster monitoring is enabled for the **openshift-file-integrity** project

16. Configure a project template

Configure your OpenShift cluster so that new projects are created with limits using the following requirements:

- The name of the limit range is **PROJECT_NAME-limits**
where **PROJECT_NAME** is the name of the project created using **oc new-project**
- The amount of memory consumed by a single container is:
 - Minimum: **128Mi**
 - Maximum: **1Gi**
 - Default: **512Mi**
 - Default request: **256Mi**

17. Create a cron job

Create a cron job with the following requirements:

- The cron job name is **ipsum**
- The cron job uses the image:
registry.domain6.example.com/library/job-runner:latest
- The cron job runs every minute
- The cron job successful job history limit is **10**
- The cron job runs in a project called **lorem**